Practical 04: Create a Fragment and Demonstrate Fragment Lifecycle

Source: Android Programming Practical(1).pdf | Source pages: 31–36

Aim

To develop and test an Android application demonstrating create a fragment and demonstrate fragment lifecycle.

Steps to Perform the Practical

1. Create a project named FragmentLifecycleDemo.

2. In Android Studio, create a blank fragment named DemoFragment and its fragment_demo layout.

3. Use the supplied DemoFragment.java code and add Log.d() statements for fragment lifecycle callbacks.

4. Add the fragment to the activity layout or use the integration method shown in the source practical.

5. Run the app and open Logcat with the FragmentLifecycle tag.

6. Observe the lifecycle messages when the fragment is attached, created, its view is created, and it is removed or the activity changes.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

2. Fragments and Activity Interaction

4. Develop an Android application to create and integrate a Fragment and demonstrate Fragment lifecycle methods.

Project Name: FragmentLifecycleDemo

Create a new fragment class:

java->new-> fragment->fragment(blank)

Fragment Name: DemoFragment

Fragment Layout name: fragment_demo

I

Filename: DemoFragment.java

```
package com.example.fragmentlifecycledemo;

import android.content.Context;

import android.os.Bundle;

import android.util.Log;

import android.view.LayoutInflater; import android.view.View;

import android.view.ViewGroup;

import androidx.annotation.NonNull; import androidx.annotation.Nullable;

import androidx.fragment.app.Fragment;

public class DemoFragment extends Fragment {

   String TAG = "FragmentLifecycle";


   @Override

   public void onAttach(@NonNull Context context) {

   super.onAttach(context);
```

```
   Log.d(TAG, "onAttach() executed");}



   @Override

  public void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       Log.d(TAG, "onCreate() executed");

   }



   @Override

   public View onCreateView(LayoutInflater inflater, ViewGroup
container,Bundle savedInstanceState) {

       Log.d(TAG, "onCreateView() executed");

       return inflater.inflate(R.layout.fragment_demo, container, false);

   }



@Override

public void onViewCreated(@NonNull View view, @Nullable Bundle
savedInstanceState) {

       super.onViewCreated(view, savedInstanceState);

        Log.d("on view created invoked", "onViewCreated() executed");

}



   @Override

   public void onStart() {

       super.onStart();

       Log.d("on start invoked", "on started");
```

```
}


@Override
public void onResume() {
    super.onResume();
    Log.d("on resume invoked", "on resume");
}


@Override
public void onPause() {
    super.onPause();
    Log.d("on pause invoked", "on pause");
}


@Override
public void onStop() {
    super.onStop();
    Log.d("on stop invoked", "on stopped");
}


@Override
public void onDestroyView() {
    super.onDestroyView();
    Log.d("on destroyview invoked", "on destroyview() executed");
}
```

```
    @Override
    public void onDestroy() {
        super.onDestroy();
        Log.d("on destroy invoked", "on destroy");
    }

    @Override
    public void onDetach() {
        super.onDetach();
        Log.d("on detach invoked", "on detach");
    }
}
```

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/fragment_container_view_tag"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
```

```
        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Hello World!"

        app:layout_constraintBottom_toBottomOf="parent"

        app:layout_constraintEnd_toEndOf="parent"

        app:layout_constraintStart_toStartOf="parent"

        app:layout_constraintTop_toTopOf="parent" />


</androidx.constraintlayout.widget.ConstraintLayout>
```

Filename: MainActivity.java

```
package com.example.fragmentlifecycledemo;


import android.os.Bundle;

import androidx.fragment.app.FragmentTransaction;

import androidx.activity.EdgeToEdge;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.graphics.Insets;

import androidx.core.view.ViewCompat;

import androidx.core.view.WindowInsetsCompat;


public class MainActivity extends AppCompatActivity {


    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);

                DemoFragment fragment = new DemoFragment();

                FragmentTransaction transaction =

                        getSupportFragmentManager().beginTransaction();

                transaction.replace(R.id.fragment_container_view_tag,
fragment);

                transaction.commit();

        }

    }
```

fragment_demo.xml

```
<?xml version="1.0" encoding="utf-8"?>

<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    tools:context=".DemoFragment">



    <!-- TODO: Update blank fragment layout -->

    <TextView

        android:layout_width="match_parent"

        android:layout_height="match_parent"

        android:text="@string/hello_blank_fragment" />



</FrameLayout>
```